

A Value Decomposition for Unforecastable Supply Disruptions:

information, routing, and inventory — and how the dominant lever flips
(deterministic study on the thesis simulator; no RL; all numbers from audited runs)

Aidan Hansen

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Summary

We completed and stress-tested the value decomposition across two regimes: the thesis scenario (one manufacturer at 5% for 48 periods — “slack,” because the surviving chain has headroom) and a documented saturated extension (the surviving manufacturer also cut). Four results. **(1) In the slack regime the binding constraint is information, not intelligence:** a two-line reroute rule triggered on the *shared upstream machine-state signal* exactly reproduces the perfect-foresight oracle (bit-identical trajectories), cutting cost from 1.06M to 295k and lost patients from 445 to 203; without that shared signal, *no* downstream policy of any sophistication can act in time, because the disrupted distributor’s own stock masks the failure from downstream observers for ~ 6 periods (~ 0.9 M of value). The measured room left for any learned policy above the deployable compound is ≈ 0 . **(2) Under scarcity the frontier becomes real and is owned by one composition:** a standing buffer *at the surviving chain* + the info-sharing reroute + an order taper achieves -79% cost at fill $0.990 - 1.000$ with the larger buffer — (urgent0) and -68% lost patients (urgent20); the parts alone achieve less than half of that and fill ≤ 0.99 . **(3) The dominant lever flips along two boundaries** — no action (disruptions shorter than the stock-cover window) $\rightarrow$ routing (slack) $\rightarrow$ compound (mild scarcity) $\rightarrow$ buffer-only (severe scarcity, where sharp rerouting becomes the *worst* measured policy). **(4) Every “clever” reactive rule we tested** (delivery-rate detection, backlog-chasing or rotating allocation, accounting fixes for the glut) **lost to a simpler alternative**, and three prior headline numbers (the $-468\text{k}/-654\text{k}$ decomposition, the “insurance captures 42%” result, and static-priority allocation’s 12% gain) were shown to be artifacts of regime or of the broken routing layer. All gates passed; both phases were independently audited by an agent re-deriving every number from raw CSVs (Phase 1: 12/12 PASS; Phase 2: 11/12, the exception being a display-rounding over-statement of one fill rate, corrected throughout this report).

Table 1: Protocol. Everything runs on the verified harness of the routing study.

Baselines	fairness-neutral (rung-c routing + proportional allocation, 1.21M) and fairness-costed (+ static priority, 1.06M), both carried through all comparisons
Seeds	pre-registered: tuning 1–10 (all parameter choices), reporting 11–30 (all reported numbers), paired; RNG-reseed fix in place (gate-guarded)
Scoring	system-level (six agents), phase-stratified; dual accounting: full cost and cost excluding manufacturer-backlog (dead-factory queue book-keeping)
Panel	fill rate, lost patients (urgent20 only, never pooled), peak backlogs, AUB, holding-vs-backlog decomposition, glut, HC dispersion, bullwhip
Saturated regime	documented scope expansion: MN_{healthy} also cut (to 70/50/30%) during the window; durations {5, 17, 48}; gates re-run per regime (saturation sanity: fill $0.875 < 1$ under routing at sat50)
Detection constraint	detection may trigger discrete pre-registered rules (reroute, order-taper) — never a continuous ordering optimizer
Audit	each phase independently re-derived from raw CSVs by a separate agent (no shared analysis code); Phase 1: 12/12 PASS

Table 2: Slack regime (urgent0 dp-cost = during+post system cost, reporting seeds; lost patients from urgent20). Full hypothesis cards in `hypotheses/`.

Policy	dp-cost	ex-MN-backlog	lost
compound baseline (routing + priority alloc)	1,062,538	—	445
+ upstream-signal reroute (h3b)	294,796	212,636	203
+ MN-down order taper (= deployable compound)	273,088	220,648	204
oracle ceiling (perfect onset + JIT pre-build)	284,120	234,485	207

1 Setup and protocol

2 Phase 1 — the slack regime: information is the binding constraint

The information result. The upstream-signal reroute — flag a source while its feeding manufacturer is below 50% of nominal lines (exactly the visibility the thesis’s full information-sharing scenario grants), step the order split away from it — reproduces the perfect-foresight oracle *bit-for-bit* (independently audited). A delivery-rate detector, by contrast, is structurally blind for ~ 6 periods (the dead distributor serves from its own stock, so its delivery statistics look healthy exactly while the pile-up begins) and ends up *worse* than the built-in trust mechanism, which is itself such a detector. The $\sim 750k$ between “no sharing” and “sharing” is therefore an *information* value, not an intelligence value — no downstream policy, learned or not, can close it. **This reframes the thesis’s information-sharing narrative: the value of sharing is real, large, and capturable by a two-line rule.**

The deployable compound now IS the measured ceiling. Adding the order taper (a discrete rule on the same shared signal: while your manufacturer is down, order only what it delivers) the deployable compound (273k) *beats* every oracle-assisted composition we built (284k+) — foresight adds nothing beyond the shared signal. Dual accounting keeps the taper honest: its gain is almost

entirely dead-factory queue bookkeeping (ex-MN it is $\approx$ neutral), and it does not move patients.

Negative results (all gated, all audited). Standing buffers in the slack regime have zero-to-negative coverage at every size (a buffered $DS_{\text{disrupted}}$ feeds the dead factory’s queue and the recovery glut) — this supersedes the earlier “permanent insurance captures 42%” finding, which was an artifact of the broken routing layer. Accounting-side glut fixes fail by construction (the double-order is committed mid-window); instant rerouting removes the glut as a side effect (peak excess 478 vs 2,967). Fairness-neutral variants of priority allocation fail: rotating priority is catastrophic (+53%, lost 740) — the trust feedback rewards *consistency*; there is no fairness-free priority.

3 Phase 2 — scarcity: the frontier becomes real

When the surviving manufacturer is also cut to 50% (supply 220 vs demand 240/period), routing still reduces cost (−26% deployable, −50% oracle) but *cannot create product*: fill pins at ~ 0.9 and lost patients barely move across the entire routing ladder (1,768 $\rightarrow$ $\sim 1,600$; in slack the same ladder achieved 203). The residual is inventory’s domain:

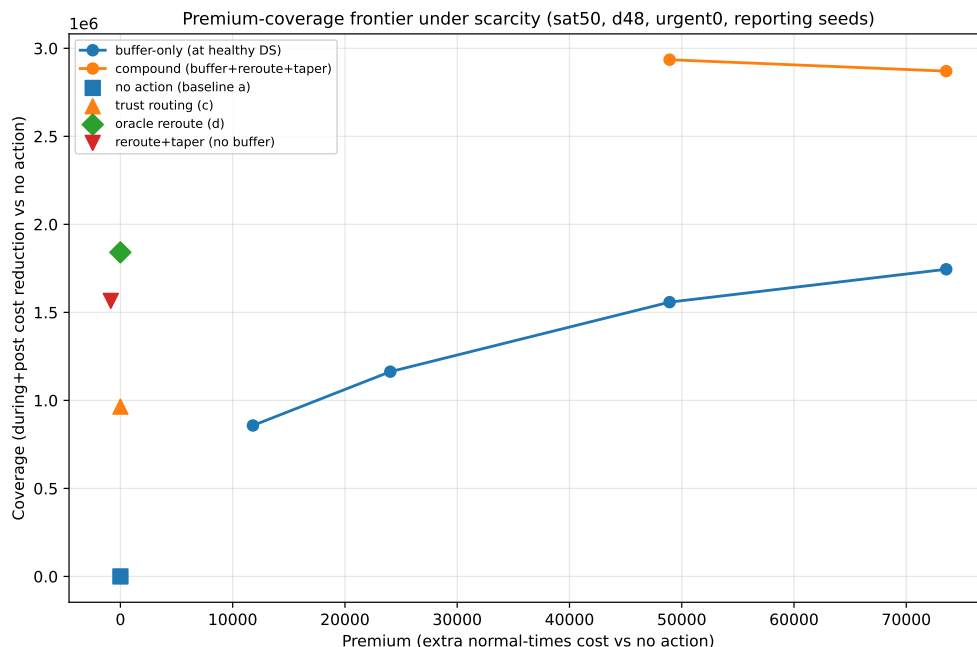


Figure 1: The premium–coverage frontier under scarcity (sat50, d48, urgent0). The composition (orange) dominates the buffer-only curve at matched premium by $\sim 2.4\times$ — the pre-registered confirmatory test passes. Compound at B960: dp-cost 766,830 (−79% vs no action), fill 0.990, premium 49k; at B1440 fill reaches 1.000. Under urgent20 (deeper effective scarcity) the larger buffer wins: lost patients 505 vs baseline 1,591 (−68%). (An earlier draft over-stated B960’s fill as 1.000 from display rounding; the independent audit caught it — exact values shown.)

Three structural findings from the screen: buffer **location** is first-order (at the surviving chain $\gg$ split $\gg$ at the disrupted chain, which is near-worthless); ordering shapers alone do nothing (supply-bound); and **static-priority allocation flips harmful under scarcity** (+23% vs proportional) — starving one health center at the queue-bound surviving distributor collapses its trust in the *healthy* chain and routes its orders toward the dead one. The slack-regime allocation gain does not transfer.

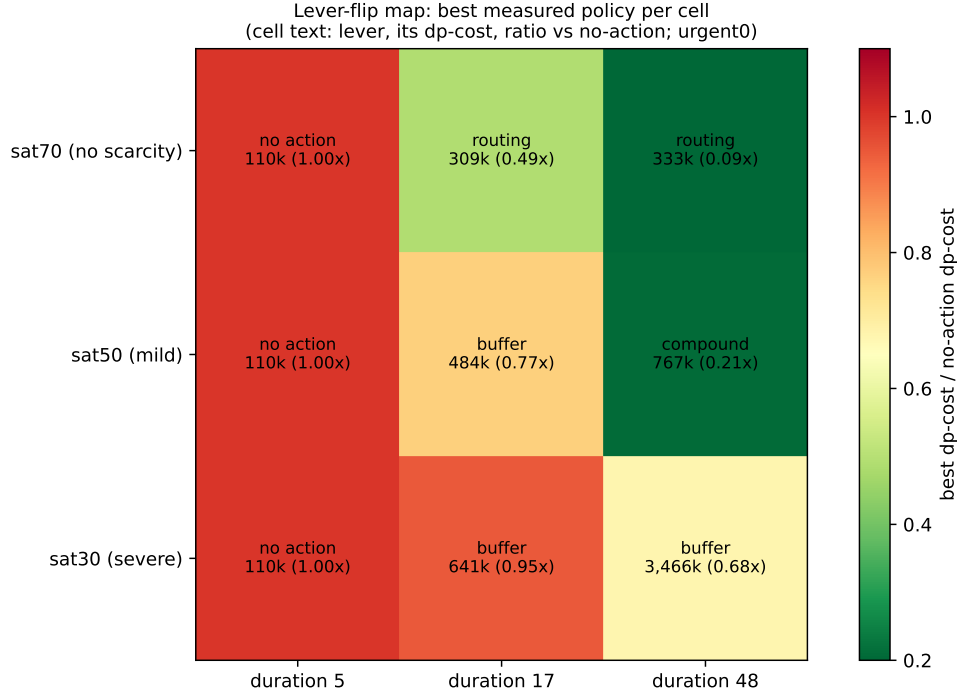


Figure 2: The lever-flip map (urgent0, reporting seeds; cell = best measured policy). Two boundaries: a *duration* boundary at the stock-cover window (~ 6 periods — for shorter disruptions every response is churn and the oracle reroute is the worst overreaction), and a *severity* boundary at the surviving chain’s headroom (at sat30, d48, sharp rerouting is the worst measured policy — 7,657k vs 5,133k for doing nothing — while a shortfall-sized buffer reaches fill 1.000 at -32%).

4 What we learned

1. **The decomposition, completed:** value under unforecastable disruption = information sharing (the largest single item in the slack regime, $\sim 750k$, capturable by a two-line rule) + routing flexibility (dominant while the surviving chain has headroom) + correctly-located inventory (necessary exactly when it doesn't) – over-reaction (responses to sub-stock-cover disruptions; sharp rerouting under severe scarcity; clever reactive allocation everywhere).
2. **The room for learned policies is ≈ 0 in the slack regime** (the deployable compound matches/beats every oracle composition) **and not visible under scarcity** (no measured policy beats the simple compound; the space closed under screen-and-compose — VNS was, per protocol, not justified: there is no residual gap for it to search).
3. **Reactive cleverness consistently backfires** via the trust feedback loop: delivery-rate detection, backlog-priority, serve-captive, rotating priority, and (under scarcity) static priority all lose to simpler alternatives. Consistency and correct *location* beat adaptivity in every measured case.
4. **Three prior numbers are formally superseded:** the $-468k/-654k$ decomposition and the 42%-insurance result (artifacts of the broken routing layer), and allocation's 12% gain (regime-specific, sign-flips under scarcity).
5. **Methodological:** the dual-accounting amendment caught a lever (the taper) whose value is bookkeeping; the audit protocol (independent re-derivation) passed both phases; two latent simulator bugs from earlier studies (RNG reseed, dead config) remained guarded by gates throughout.

5 Next steps

1. **Severity-aware redirect** (new policy class — awaiting your approval, spec in `NEXT_PLAN.md`): cap rerouted volume at the surviving chain's observed headroom. Hypothesis: removes the only failure mode of the compound (over-redirection at severe scarcity), making one policy safe across the whole map.
2. **Demand-aware buffer sizing** (B from measured demand minus surviving capacity — the urgent20 result shows the frozen urgent0-scaled grid undersizes when demand grows).
3. **Recurring-disruption axis** (new regime — your call): standing insurance amortizes across events; a single-event episode understates it.
4. **For the eventual corrected DRL:** the bar is the compound, per regime — 273k (slack) / 767k–1,865k (sat50, by config) — same seeds, same gates, both accounting conventions, fairness-preserving. Given items 1–3, the honest expectation remains that simple rules saturate this environment; the DRL comparison should be framed as measuring *how close* learning comes to the structured optimum, not whether it “wins.”
5. **Paper framing** (consistent with the advisor's decision-first agenda): a value-of-information and value-of-structure decomposition with a lever-flip map, in which learned policies have measured, near-zero residual room — a cost-of-AI result grounded entirely in audited, deterministic experiments.